
VChO – 2026

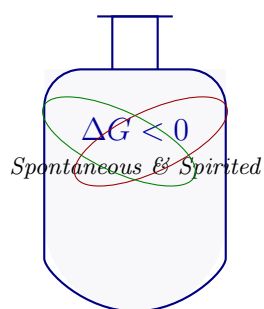
Vedantu Chemistry Olympiad

The Gibbs Free Thinkers

Sarvagya Gangwar

Subhadeep Panda

Verit



Contents

1	Use of Iron in “Blue” Colours	4
1.1	Oxalate Acidic Permanganate Titration	4
1.2	Powdered Zinc Reduction Processes	4
1.3	Iron Content Back-Titration Half-Cells	4
1.4	Quantitative Analytical Computations	5
1.5	Structural Photolytic Fragmentation Cascade	6
1.6	Turnbull’s Blue Coordinate Recovery Matrix	6
2	Quantum Transitions and Probability Distributions	7
2.1	Frequency of the Lowest Energy Transition	7
2.2	Identification of UV Lines and Determination of C_0	8
2.3	Classical Velocity Equation from Bohr Postulates	9
2.4	Most Probable Radius and Maximum Density value	9
2.5	Realistic Discrepancies of the Bohr Model	9
2.6	Hydrogenic Probability Shifts	10
2.7	Radiative Decay Lifetime Analysis	10
2.8	Color Characterization of De-excitation Emission	11
3	Light Absorption and Energy Conversion	12
3.1	Fraction of Light Absorbed by the Sample	12
3.2	Photon Capture Rate from the Laser Source	13
3.3	Molecular Absorption Cross-Section	13
3.4	Energy Storage Efficiency in Photosynthesis	14
4	Schrödinger Equation	15
4.1	Part A: Particle in a One-Dimensional Box	15
4.1.1	Schrödinger Equation Form	15
4.1.2	Wave Functions and Shape	15
4.1.3	Derivation of Energy Levels	15
4.2	Part B: Particle in a 2-D Box	17
4.2.1	Multi-Electron Configuration Table	17
4.2.2	Identification of the Highest Filled State	17
4.2.3	Magnetic Properties	17

4.2.4	Light Absorption Wavelength	18
5	Problem 5: Laser Cooling Mechanics	19
5.1	Temperature and Speed of the Gas Atoms	19
5.2	Properties of the Incoming Laser Light	19
5.3	Slowing the Atoms via Photon Collisions	20
5.4	Electron Ground State Configurations	20
5.5	Electron Excited State Configurations	20
5.6	Total Momentum Options	20
5.7	Energy Level Splitting	21

1 Use of Iron in “Blue” Colours

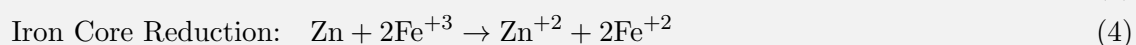
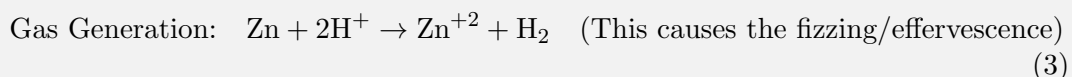
1.1 Oxalate Acidic Permanganate Titration

Here are the separate oxidation and reduction half-reactions for balancing the titration under acidic conditions:



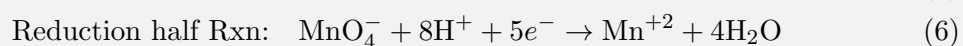
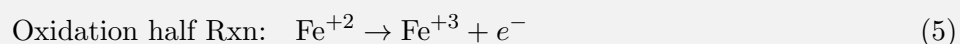
1.2 Powdered Zinc Reduction Processes

When we add excess zinc powder, it reduces both the acid protons (which causes bubbling) and the iron center ions:



1.3 Iron Content Back-Titration Half-Cells

Next, we titrate the newly formed iron ions with potassium permanganate. The half-reactions for this step are:



1.4 Quantitative Analytical Computations

Scenario I: Oxalate Species Calculation

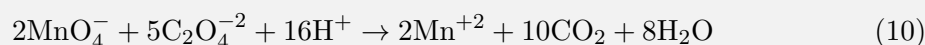
First, let's write down the concentration and volume numbers we have for KMnO_4 :

$$M_{\text{KMnO}_4} = 0.0196 \text{ M} \quad (7)$$

$$V_{\text{KMnO}_4} = 0.0311 \text{ L} \quad (8)$$

$$n_{\text{KMnO}_4} = 0.0196 \times 0.0311 = 6 \times 10^{-4} \text{ mol} \quad (9)$$

By combining the two half-reactions from part 1.1 so that the electrons (e^-) cancel out, we get the overall balanced equation:



Using the mole ratios from this equation:

$$\frac{2}{n_{\text{MnO}_4^-}} = \frac{5}{n_{\text{C}_2\text{O}_4^{2-}}} \quad (11)$$

$$\Rightarrow n_{\text{C}_2\text{O}_4^{2-}} = (2.5) \times (6 \times 10^{-4}) = \mathbf{1.5 \times 10^{-3} \text{ mol}} \quad (12)$$

Scenario II: Iron Content Determination

Now for the second titration run after using zinc, the numbers are:

$$\text{Volume of KMnO}_4 = 0.0051 \text{ L} \quad (13)$$

$$\text{Moles of KMnO}_4 = 0.0196 \times 0.0051 = 9.996 \times 10^{-5} \text{ mol} \quad (14)$$

Combining the half-reactions from part 1.3 shows that 5 mol of Fe^{+2} reacts with every 1 mol of MnO_4^- . So:

$$\text{Moles of Fe}^{+2} = 5 \times (9.996 \times 10^{-5}) = \mathbf{4.998 \times 10^{-4} \text{ mol}} \quad (15)$$

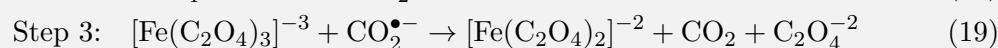
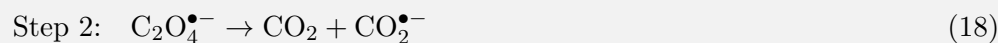
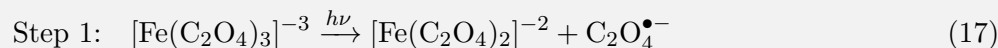
Scenario III: Relative Stoichiometric Core Ratio

Let's find the ratio between the iron and the oxalate moles to see what the compound looks like:

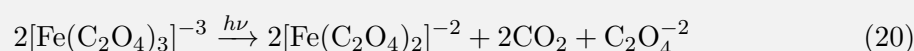
$$\frac{\text{Moles of Fe}^{+2}}{\text{Moles of C}_2\text{O}_4^{2-}} = \frac{4.998 \times 10^{-4}}{1.5 \times 10^{-3}} = 0.33 \approx \frac{1}{3} \quad (16)$$

1.5 Structural Photolytic Fragmentation Cascade

Here are the elementary steps that show how the complex breaks down into free radicals when light hits it:



If we add steps 1, 2, and 3 together, we get the total photochemical reaction equation:



Now we can calculate the overall quantum yield (ϕ):

$$\phi = \frac{\text{Moles of Fe}^{+2} \text{ made}}{\text{Moles of photons } (h\nu) \text{ absorbed}} = \frac{2}{1} \implies \phi = 2 \quad (21)$$

1.6 Turnbull's Blue Coordinate Recovery Matrix

(i) Matrix Displacement Balance:



(ii) Product Yield Quantification via Atom Conservation (POAC):

We can use the Principle of Atom Conservation on our 5.1 mL titration data. Since there are 2 iron atoms inside each formula unit of the product complex, the final mass calculations work out like this:

$$n_{\text{KFe}[\text{Fe}(\text{CN})_6]} = \frac{4.998 \times 10^{-4}}{2} \quad (23)$$

$$\text{Mass of KFe}[\text{Fe}(\text{CN})_6] = \frac{4.998 \times 10^{-4}}{2} \times 306.84 = \mathbf{0.076 \text{ g}} \quad (24)$$

2 Quantum Transitions and Probability Distributions

2.1 Frequency of the Lowest Energy Transition

The lowest energy transition corresponds to the maximum wavelength ($\lambda_{\max}$). Given:

$$\lambda_{\max} = 656.3 \times 10^{-9} \text{ m} \quad (25)$$

$$c = 3 \times 10^8 \text{ m/s} \quad (26)$$

Evaluating the frequency f :

$$f = \frac{c}{\lambda} = 4.57 \times 10^{14} \text{ Hz} \quad (27)$$

2.2 Identification of UV Lines and Determination of C_0

Energy levels are modeled according to $E_n = -C_0/n^2$. The energy difference between two levels is:

$$E_{n_1} - E_{n_2} = -C_0 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (28)$$

$$\implies \frac{hc}{\lambda} = C_0 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (29)$$

In Spectrum 1 only, we can see wavelengths that belong to the visible range. From there, we transition right into the ultraviolet (UV) region, which lets us identify it as the Lyman series (since λ decreases and E increases, meaning $E_{(n_2 \rightarrow n_1)} > E_{(n_3 \rightarrow n_2)}$).

For the first line after Spectrum 1 in the Lyman series:

$$n_1 = 1, \quad n_2 = 2 \implies \lambda = 121.6 \text{ nm} \quad (30)$$

Plugging these integers into the energy balance equation:

$$\frac{hc}{\lambda} = C_0 \left(\frac{1}{1} - \frac{1}{4} \right) \quad (31)$$

$$\implies \frac{hc}{\lambda} = \frac{3C_0}{4} \quad (32)$$

$$\implies C_0 = \frac{4hc}{3\lambda} = \frac{4 \times 6.626 \times 10^{-34} \text{ J} \cdot \text{s} \times 3 \times 10^8 \text{ m/s}}{3 \times 121.6 \times 10^{-9} \text{ m}} \quad (33)$$

$$\implies C_0 = \mathbf{2.18 \times 10^{-18} \text{ J}} \quad (34)$$

Let's find this constant in terms of wavenumber (cm^{-1}):

$$E = \frac{hc}{\lambda} \implies \frac{1}{\lambda} = \frac{E}{hc} \quad (35)$$

$$\frac{1}{\lambda} = \frac{2.18 \times 10^{-18} \text{ J}}{6.626 \times 10^{-34} \text{ J} \cdot \text{s} \times 3 \times 10^8 \text{ m/s}} \quad (36)$$

$$\frac{1}{\lambda} = 1.09 \times 10^7 \text{ m}^{-1} = \mathbf{109,737 \text{ cm}^{-1}} \quad (37)$$

2.3 Classical Velocity Equation from Bohr Postulates

Balancing centripetal force with the electrostatic Coulomb force gives us:

$$\frac{mv^2}{r} = \frac{kZe^2}{r^2} \quad (38)$$

Using Bohr's angular momentum quantization rule:

$$mvr = \frac{nh}{2\pi} \implies r = \frac{nh}{2\pi mv} \quad (39)$$

Substituting r back into the force balance equation:

$$\frac{nh}{2\pi mv} \cdot mv^2 = kZe^2 \quad (40)$$

$$\implies v = \frac{2\pi kZe^2}{nh} \quad (41)$$

Using the permittivity constant value $k = \frac{1}{4\pi\epsilon_0}$:

$$v = \frac{2\pi Ze^2}{4\pi\epsilon_0 nh} = \frac{Ze^2}{2\epsilon_0 nh} = 2.18 \times 10^6 \times \left(\frac{Z}{n}\right) \text{ m/s} \quad (42)$$

2.4 Most Probable Radius and Maximum Density value

$$f(r) = \frac{4Z^3}{a_0^3} r^2 e^{-\frac{2Zr}{a_0}} \implies f(r) = \frac{4}{a_0^3} r^2 e^{-\frac{2r}{a_0}} \quad (43)$$

To find the most probable radius where we can find the electron, we take the derivative and set $f'(r) = 0$:

$$\frac{4}{a_0^3} \left[2r e^{-\frac{2r}{a_0}} + r^2 e^{-\frac{2r}{a_0}} \cdot \left(-\frac{2}{a_0}\right) \right] = 0 \quad (44)$$

$$1 + r \left(-\frac{1}{a_0}\right) = 0 \quad (45)$$

$$\implies r = a_0 \quad (\text{Most probable value of } r \text{ for H atom}) \quad (46)$$

2.5 Realistic Discrepancies of the Bohr Model

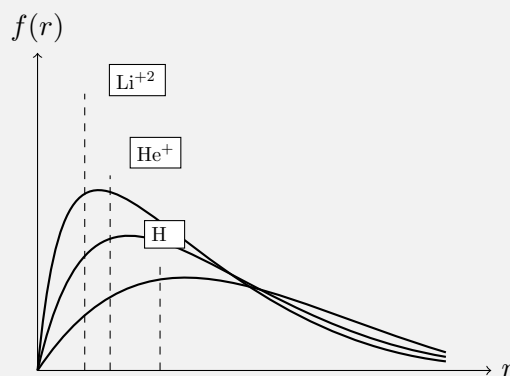
- a) The Bohr model treats electrons like tiny planets moving in fixed, predictable orbits. In reality, this is impossible because the Heisenberg Uncertainty Principle says we cannot know an electron's exact path and position at the same time.

2.6 Hydrogenic Probability Shifts

Differentiating the general distribution function gives us the location of the maximum peak:

$$f(r) = \frac{4Z^3}{a_0^3} r^2 e^{-\frac{2Zr}{a_0}} \xrightarrow{f'(r)=0} r_{\max} = \frac{a_0}{Z} \quad (47)$$

As the nuclear charge Z increases, the peak of the curve shifts closer and closer to the nucleus ($r_{\max}$ gets smaller).



2.7 Radiative Decay Lifetime Analysis

The decay of light intensity follows standard first-order kinetics:

$$-\frac{dI}{dt} = kI \implies \int_{I_0}^I -\frac{dI}{I} = \int_0^t k \cdot dt \quad (48)$$

$$\implies [\ln I]_{I_0}^I = -kt \quad (49)$$

$$\implies \ln \left(\frac{I}{I_0} \right) = -kt \implies I = I_0 e^{-kt} \quad (50)$$

Using the given boundaries $I_0 = 100$, $I = 10$, and the rate constant $k = 10^8 \text{ s}^{-1}$:

$$10 = 100e^{-kt} \quad (51)$$

$$\frac{1}{10} = e^{-kt} \quad (52)$$

$$\ln \left(\frac{1}{10} \right) = -10^8 t \quad (53)$$

$$\ln 10 = 10^8 t \quad (54)$$

$$\implies t = \frac{\ln 10}{10^8} = 2.303 \times 10^{-8} \text{ sec} \quad (55)$$

2.8 Color Characterization of De-excitation Emission

Because the electron drops from an excited state straight down to the ground state ($n_2 = 2 \rightarrow n_1 = 1$), this transition falls completely within the **Lyman Region**. Since the Lyman series is entirely in the ultraviolet spectrum, **no visible color** will be seen by the human eye.

3 Light Absorption and Energy Conversion

3.1 Fraction of Light Absorbed by the Sample

We are given the following values for the sample:

$$c = 1.35 \times 10^{-6} \text{ M} \quad (56)$$

$$\varepsilon = 1.42 \times 10^5 \text{ M}^{-1}\text{cm}^{-1} \quad (57)$$

$$b = 1.00 \text{ cm} \quad (58)$$

First, let's calculate the absorbance (A) using the Beer-Lambert law:

$$A = \varepsilon \cdot c \cdot b \quad (59)$$

$$A = (1.42 \times 10^5) \times (1.35 \times 10^{-6}) \times 1.00 \quad (60)$$

$$A = 0.1917 \quad (61)$$

Now, we know that absorbance relates to transmittance (T) by the formula $A = -\log_{10}(T)$. So we can find T :

$$T = 10^{-A} \quad (62)$$

$$T = 10^{-0.1917} = 0.6431 \quad (63)$$

The transmittance T is the fraction of light that passes through (I/I_0). Therefore, the fraction of light that actually gets absorbed (α) is:

$$\alpha = 1 - T = 1 - 0.6431 = 0.3569 \quad (64)$$

So, about **35.7%** of the incident light is absorbed by the sample.

3.2 Photon Capture Rate from the Laser Source

We have a laser with a power of $P = 0.15 \text{ mW} = 0.15 \times 10^{-3} \text{ W}$ (which is J/s), operating at a wavelength of $\lambda = 680 \text{ nm} = 680 \times 10^{-9} \text{ m}$.

First, let's find the energy (E) of just one single photon:

$$E = \frac{hc}{\lambda} \quad (65)$$

$$E = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s}) \times (3.00 \times 10^8 \text{ m/s})}{680 \times 10^{-9} \text{ m}} \quad (66)$$

$$E = 2.923 \times 10^{-19} \text{ J} \quad (67)$$

Next, we figure out the total number of photons hitting the sample per second (N_{total}):

$$N_{\text{total}} = \frac{P}{E} = \frac{0.15 \times 10^{-3} \text{ J/s}}{2.923 \times 10^{-19} \text{ J}} = 5.132 \times 10^{14} \text{ photons/s} \quad (68)$$

Since we found out in part 4.1 that only 35.69% of the light gets absorbed, the actual photon capture rate (R_{abs}) is:

$$R_{\text{abs}} = \alpha \times N_{\text{total}} \quad (69)$$

$$R_{\text{abs}} = 0.3569 \times 5.132 \times 10^{14} \quad (70)$$

$$R_{\text{abs}} = \mathbf{1.83 \times 10^{14} \text{ photons/s}} \quad (71)$$

3.3 Molecular Absorption Cross-Section

The molecular absorption cross-section (σ) is basically the effective area one molecule covers to capture a photon. We can convert the molar extinction coefficient ε to σ using Avogadro's number ($N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$) and converting liters to cm^3 :

$$\sigma = \frac{\ln(10) \times \varepsilon}{N_A \times 10^3} \quad (72)$$

Let's plug in our value of $\varepsilon = 1.42 \times 10^5 \text{ M}^{-1}\text{cm}^{-1}$:

$$\sigma = \frac{2.3026 \times (1.42 \times 10^5)}{6.022 \times 10^{23} \times 10^3} \quad (73)$$

$$\sigma = \frac{326969.2}{6.022 \times 10^{26}} \quad (74)$$

$$\sigma = \mathbf{5.43 \times 10^{-22} \text{ cm}^2/\text{molecule}} \quad (75)$$

(Or $5.43 \times 10^{-26} \text{ m}^2/\text{molecule}$ if written in SI units).

3.4 Energy Storage Efficiency in Photosynthesis

We are told that storing 1 mol of O_2 takes 8 mol of photons at 680 nm, and this stores $\Delta H = 467$ kJ of chemical energy.

First, let's find out how much total energy is brought in by those 8 mol of photons. The energy of 1 mol of photons (an Einstein) at this wavelength is:

$$E_{\text{mole}} = N_A \times \frac{hc}{\lambda} \quad (76)$$

$$E_{\text{mole}} = (6.022 \times 10^{23}) \times (2.923 \times 10^{-19} \text{ J}) \quad (77)$$

$$E_{\text{mole}} = 1.760 \times 10^5 \text{ J/mol} = 176.0 \text{ kJ/mol} \quad (78)$$

So the total energy supplied by 8 mol of photons is:

$$E_{\text{total}} = 8 \times 176.0 \text{ kJ} = 1408.0 \text{ kJ} \quad (79)$$

Now, we can find the thermodynamic conversion efficiency (η) by comparing the stored energy to the input energy:

$$\eta = \frac{\Delta H}{E_{\text{total}}} \times 100\% \quad (80)$$

$$\eta = \frac{467 \text{ kJ}}{1408.0 \text{ kJ}} \times 100\% \quad (81)$$

$$\eta = \mathbf{33.17\%} \quad (82)$$

4 Schrödinger Equation

4.1 Part A: Particle in a One-Dimensional Box

4.1.1 Schrödinger Equation Form

Inside the one-dimensional box where the potential energy is zero ($V(x) = 0$), the wave function satisfies this differential equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x) \quad \Rightarrow \quad \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2mE}{h^2}\psi(x) = 0 \quad (83)$$

4.1.2 Wave Functions and Shape

When we apply the boundary conditions where the wave function must be zero at the walls ($\psi(0) = 0$ and $\psi(L) = 0$), only the sine wave solutions work:

$$\psi(x) = \sin\left(\frac{n\pi x}{L}\right) \quad (84)$$

4.1.3 Derivation of Energy Levels

Taking the second derivative of our wave function gives:

$$\frac{d^2}{dx^2} \left[\sin\left(\frac{n\pi x}{L}\right) \right] = -\left(\frac{n\pi}{L}\right)^2 \sin\left(\frac{n\pi x}{L}\right) \quad (85)$$

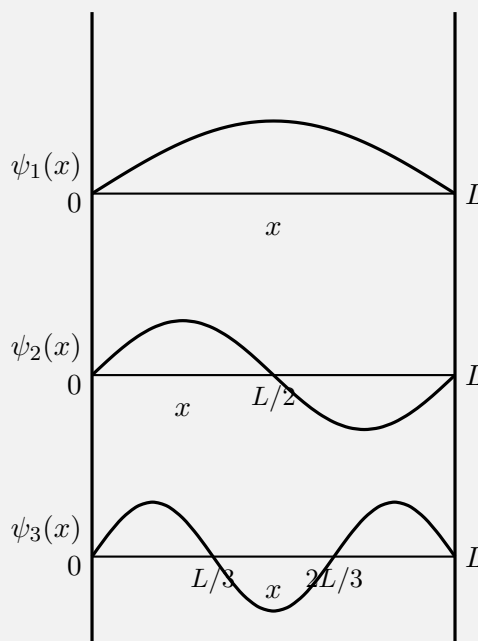
Plugging this result back into our main differential equation allows us to solve for the energy (E_n):

$$-\left(\frac{n\pi}{L}\right)^2 \psi(x) + \frac{8\pi^2mE}{h^2}\psi(x) = 0 \quad \Rightarrow \quad E_n = \frac{h^2n^2}{8mL^2} \quad (86)$$

4.1.4 Node Analysis & Schematic Plot

The number of internal nodes within a one-dimensional box (locations where the wave function crosses zero, excluding the boundary walls) is given by:

$$\text{Number of Nodes} = n - 1 \quad (87)$$



- **Ground State ($n = 1$):** Contains **0 nodes**. The wave function reaches its peak amplitude right in the middle of the box ($x = L/2$).
- **First Excited State ($n = 2$):** Contains **1 node**. The wave profile crosses the zero baseline exactly at the center ($x = L/2$).
- **Second Excited State ($n = 3$):** Contains **2 nodes**. The wave profile crosses the baseline at $x = L/3$ and $x = 2L/3$.

4.1.5 Normalization Proof

To ensure the total probability of finding the electron somewhere inside the box is exactly 1 (100%), we solve the normalization integral to find the amplitude constant A :

$$\int_0^L A^2 \sin^2 \left(\frac{\pi x}{L} \right) dx = 1 \implies A^2 \left(\frac{L}{2} \right) = 1 \implies A = \sqrt{\frac{2}{L}} \quad (88)$$

So, the fully normalized wave function for the lowest state is: $\psi_1(x) = \sqrt{\frac{2}{L}} \sin \left(\frac{\pi x}{L} \right)$.

4.2 Part B: Particle in a 2-D Box

4.2.1 Multi-Electron Configuration Table

The total energy of an electron in a 2-D box depends on the sum of squares ($n_x^2 + n_y^2$). Sorting these levels from lowest to highest energy allows us to map the complete shell occupancy:

Energy Level ($n_x^2 + n_y^2$)	Combos (n_x, n_y)	Number of Orbitals	Electrons Added
2	(1,1)	1	2
5	(1,2), (2,1)	2	4
8	(2,2)	1	2
10	(1,3), (3,1)	2	4
13	(2,3), (3,2)	2	4
17	(1,4), (4,1)	2	4
18	(3,3)	1	2
20	(2,4), (4,2)	2	4
25	(3,4), (4,3)	2	4
26	(1,5), (5,1)	2	4

4.2.2 Identification of the Highest Filled State

By tracking our total allocation of 26 electrons from the bottom up through the states, the highest energy level that holds any electrons is level 20. This level consists of two completely separate spatial states, (2, 4) and (4, 2), which share the exact same energy and hold a total of **4 electrons**.

4.2.3 Magnetic Properties

Because energy level 20 is completely full with its 4 electrons, all electron spins are paired up perfectly with opposite partners ($\uparrow\downarrow$). Since there are absolutely zero unpaired electrons left over anywhere in the box system:

Magnetic State: Diamagnetic.

4.2.4 Light Absorption Wavelength

The lowest energy light transition happens when an electron jumps from the highest filled energy level (Level 20) to the next empty level (Level 25):

$$\Delta E = \frac{h^2}{8m_e L^2} (25 - 20) = \frac{5h^2}{8m_e L^2} \quad (89)$$

Using the box width $L = 1 \text{ nm} = 10^{-9} \text{ m}$:

$$\Delta E = \frac{5 \times (6.63 \times 10^{-34})^2}{8 \times (9.11 \times 10^{-31}) \times (10^{-9})^2} = 3.016 \times 10^{-19} \text{ J} \quad (90)$$

Now we find the actual wavelength (λ_{max}) of the light needed to make this electron jump:

$$\lambda_{\text{max}} = \frac{h \cdot c}{\Delta E} = \frac{(6.63 \times 10^{-34}) \times (3.00 \times 10^8)}{3.016 \times 10^{-19}} = 6.595 \times 10^{-7} \text{ m} \approx \mathbf{660 \text{ nm}} \quad (91)$$

5 Problem 5: Laser Cooling Mechanics

5.1 Temperature and Speed of the Gas Atoms

The oven temperature is $T = 600^\circ\text{C} = 873.15\text{ K}$. The average kinetic energy of a single gas atom is calculated by:

$$E_k = \frac{3}{2}k_B T = 1.5 \times (1.3806 \times 10^{-23} \text{ J} \cdot \text{K}^{-1}) \times 873.15 \text{ K} = \mathbf{1.81 \times 10^{-20} \text{ J}}$$

The actual mass of a single ^{40}Ca atom is:

$$m = \frac{39.96 \times 10^{-3} \text{ kg} \cdot \text{mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}} = 6.636 \times 10^{-26} \text{ kg}$$

Using these values, we can find its momentum and typical thermal speed:

$$v_{\text{rms}} = \sqrt{\frac{2 \times (1.81 \times 10^{-20})}{6.636 \times 10^{-26}}} = \mathbf{738.7 \text{ m} \cdot \text{s}^{-1}}$$
$$p_{\text{rms}} = m \times v_{\text{rms}} = (6.636 \times 10^{-26}) \times (738.7) = \mathbf{4.9 \times 10^{-23} \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}}$$

5.2 Properties of the Incoming Laser Light

For a laser beam with a wavelength of $\lambda = 396.96 \text{ nm}$ ($396.96 \times 10^{-9} \text{ m}$):

$$\nu = \frac{c}{\lambda} = \frac{3.00 \times 10^8}{396.96 \times 10^{-9}} = \mathbf{7.56 \times 10^{14} \text{ Hz}}$$
$$E = h\nu = (6.626 \times 10^{-34}) \times (7.56 \times 10^{14}) = \mathbf{5.00 \times 10^{-19} \text{ J}}$$
$$p_{\text{photon}} = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34}}{396.96 \times 10^{-9}} = \mathbf{1.67 \times 10^{-27} \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}}$$

5.3 Slowing the Atoms via Photon Collisions

When an ion re-emits a photon, it shoots out in a random direction. Over many repetitions, these random kicks cancel each other out to zero. Therefore, the net slowdown comes entirely from absorbing the incoming photons:

$$\Delta p_{\text{ion}} = -p_{\text{photon}} = -1.67 \times 10^{-27} \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}$$

$$\Delta v_{\text{ion}} = \frac{\Delta p_{\text{ion}}}{m} = \frac{-1.67 \times 10^{-27}}{6.636 \times 10^{-26}} = -0.0252 \text{ m} \cdot \text{s}^{-1}$$

To bring a typical fast-moving atom completely to a stop from its initial speed (v_{rms}):

$$\text{Total Absorption Cycles Needed} = \frac{v_{\text{rms}}}{|\Delta v_{\text{ion}}|} = \frac{738.7}{0.0252} \approx \mathbf{29,313} \text{ cycles}$$

5.4 Electron Ground State Configurations

Ground State Electron Setup (Ca^+): $[\text{Ar}]4s^1$

For the single lone electron in the outermost s -subshell ($l = 0$, $s = 1/2$):

$$L = \sqrt{l(l+1)}\hbar = \mathbf{0}, \quad S = \sqrt{s(s+1)}\hbar = \sqrt{\frac{1}{2}\left(\frac{3}{2}\right)}\hbar = \frac{\sqrt{3}}{2}\hbar$$

5.5 Electron Excited State Configurations

Excited State Electron Setup (Ca^+): $[\text{Ar}]4p^1$

When the electron jumps up to a p -subshell ($l = 1$, $s = 1/2$):

$$L = \sqrt{1(2)}\hbar = \sqrt{2}\hbar, \quad S = \frac{\sqrt{3}}{2}\hbar$$

5.6 Total Momentum Options

The total momentum quantum number j combines the electron's orbital and spin properties, ranging from $|l - s|$ to $|l + s|$ in individual steps:

$$j = \left|1 - \frac{1}{2}\right|, \dots, \left|1 + \frac{1}{2}\right| \implies \mathbf{j = \frac{1}{2}, \quad j = \frac{3}{2}}$$

5.7 Energy Level Splitting

The cooling laser points at the lower split state ($j = 1/2$ at $\lambda_1 = 396.96$ nm), while the jump to the upper state ($j = 3/2$) requires a slightly different color light ($\lambda_2 = 393.48$ nm). The exact energy gap between these two split states is calculated by:

$$\Delta E_{\text{split}} = h \cdot c_0 \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)$$

$$\Delta E_{\text{split}} = (6.626 \times 10^{-34}) \times (3.00 \times 10^8) \times \left(\frac{1}{393.48 \times 10^{-9}} - \frac{1}{396.96 \times 10^{-9}} \right) = 4.43 \times 10^{-21} \text{ J}$$